Evaluation of Six Microcontrollers

Lifetime, parallel reliability, and processing speed

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STAT 4714 – Professor Courtney Kyger

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Problem and analysis plan

- Team must choose 2 microcontrollers for an autonomous vehicle based on stats findings.
- There were 6 suppliers & 12 observations per supplier in each dataset.
- Team must compare thermal cycle lifetime (in hours) and clock cycle time in milliseconds.
- Summarize all suppliers and simulate all 63 nonempty parallel combinations in MATLAB.
- Compare 15 pairs and 6 clock cycle groups using the class ANOVA and t-test methods learned from class at $\alpha = 0.05$.

Time to failure summary

Supplier	Mean	Median	Sample SD
1	4976.8439	2323.1850	5726.8839
2	6917.2299	5552.3375	5013.6839
3	4614.4611	3462.0468	4459.9591
4	3634.0783	3380.0566	1948.6646
5	7319.8507	3423.9136	7375.5191
6	3923.2351	4472.9338	2801.8157

All values in hours; $n = 12$ per supplier.

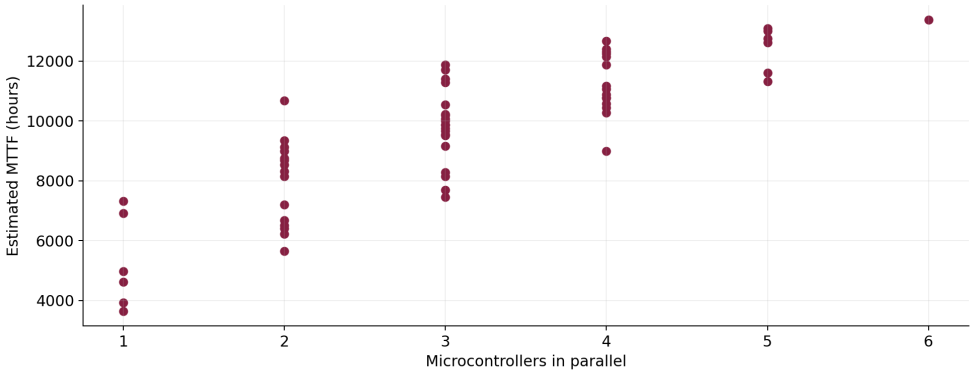
Parallel lifetime model

$$\widehat{\lambda}_j = 1/\overline{t}_j, \qquad T_j = -\log(U)/\widehat{\lambda}_j, \qquad T_{\parallel} = \max(T_1, \ldots, T_k).$$

- Independent exponential component lifetimes, same as in Milestone 2.
- Show 0 repairs, common failures, and/or accelerated failure after a component fails.
- 100,000 Monte Carlo trials; individual MTTFs use observed means too.
- $6 + 15 + 20 + 15 + 6 + 1 = 63$ supplier combinations.

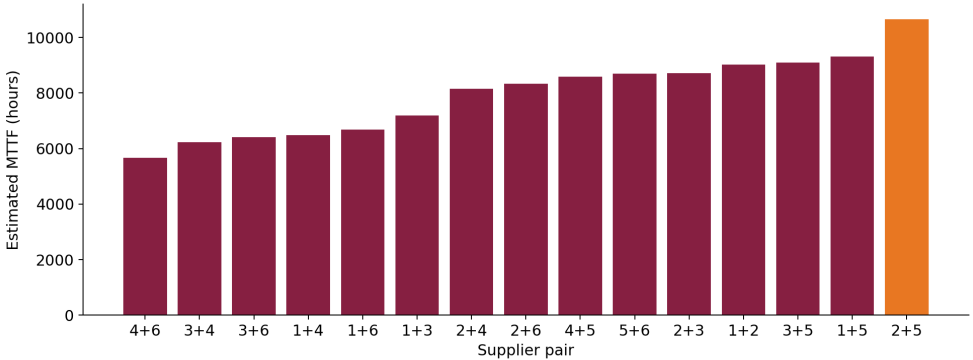
These are solely model assumptions. 12 observations per supplier do not establish a credible, exponential distribution.

More || (parallel) components increased modeled lifetime



Each point is one combination. Adding a component cannot shorten the modeled system lifetime.

Comparing all fifteen supplier pairs



Pair 2+5 has the largest modeled mean lifetime, comes to about 10,700 hours.

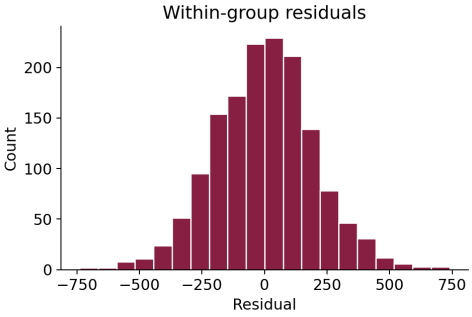
How the pair comparison was constructed

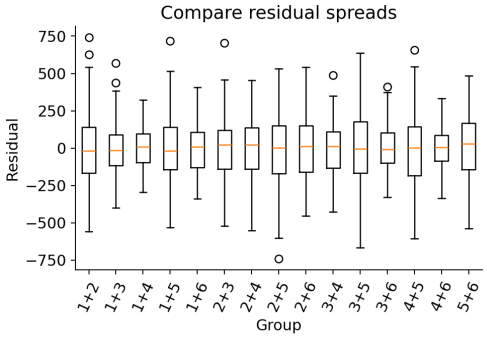
- For each pair, the team simulated 100 independent batches of 1,000 parallel lifetimes.
- Each observation for ANOVA is only one batch mean. Averaging reduces the strong skew of individual lifetimes (CLT).
- Team used fresh random draws per every and each pair to avoid dependence from share components.
- H_0: through this, all 15 fitted model pair mean lifetimes are equal.

Acknowledging scope of the test

Simulation compares fitted models. More simulated trials reduce Monte Carlo error, and it is healthy to note that they do not increase the original sample of 12 devices per supplier.

Pair ANOVA: check the assumptions





Batch residuals are roughly centered and bell shaped - spreads differ (largest/smallest variance = 3.12). Equal variance is questionable here.

Pair tests: conditional model results

$$F(14, 1485) = 4958.40, \qquad p < 10^{-16}.$$

- The classical ANOVA rejects equal fitted model pair means, subject to the variance concern.
- Follow up two sided, pooled two sample t-tests compare all 105 pair combinations (full results in a CSV).
- These are unadjusted exploratory tests, not simultaneous proof of a unique winner.
- Pair 2+5 is the numerical model lead, but hardly establishes population superiority from the original measurements.

Pair 2+5 follow up comparisons (1/2)

Other pair	t	p (unadjusted)	Decision
1+2	48.22	$< 10^{-16}$	Reject H_0
1+3	115.95	$< 10^{-16}$	Reject H_0
1+4	145.80	$< 10^{-16}$	Reject H_0
1+5	39.75	$< 10^{-16}$	Reject H_0
1+6	134.16	$< 10^{-16}$	Reject H_0
2+3	61.22	$< 10^{-16}$	Reject H_0
2+4	79.35	$< 10^{-16}$	Reject H_0

Compare 2+5 with each other pair. $df = 198$. Model only, where equal variance caveat apply.

Pair 2+5 follow up comparisons (2/2)

Other pair	t	p (unadjusted)	Decision
2+6	71.72	$< 10^{-16}$	Reject H_0
3+4	152.65	$< 10^{-16}$	Reject H_0
3+5	46.97	$< 10^{-16}$	Reject H_0
3+6	148.57	$< 10^{-16}$	Reject H_0
4+5	59.27	$< 10^{-16}$	Reject H_0
4+6	177.29	$< 10^{-16}$	Reject H_0
5+6	58.49	$< 10^{-16}$	Reject H_0

Compare 2+5 with each other pair. $df = 198$. Model only, where equal variance caveat apply

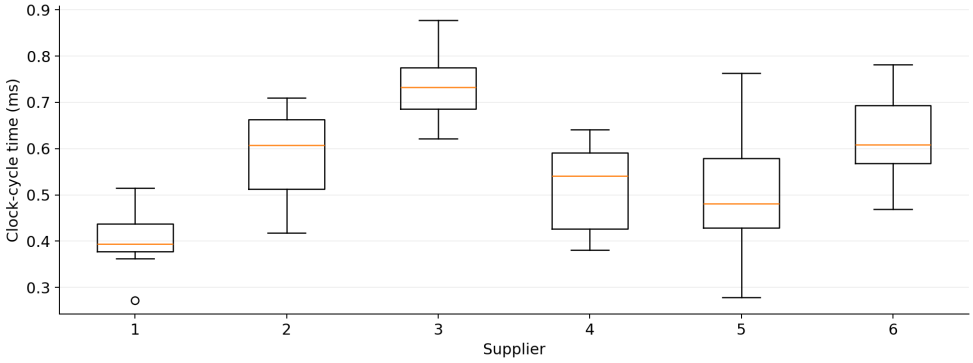
Clock cycle summary

Supplier	Mean	Median	Sample SD
1	0.4028	0.3941	0.0621
2	0.5910	0.6078	0.0934
3	0.7395	0.7322	0.0785
4	0.5170	0.5412	0.0953
5	0.5006	0.4809	0.1362
6	0.6268	0.6089	0.0990

All values in milliseconds and $n = 12$ per supplier.

Smaller clock cycle time means faster processing.

Clock cycle distributions



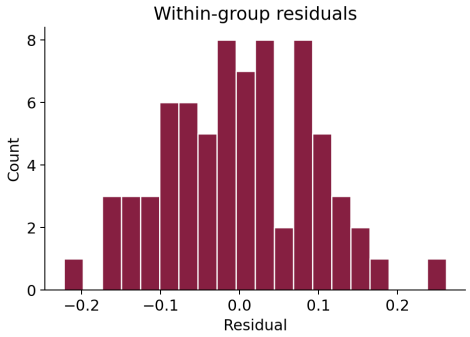
Supplier 1 has the smallest sample mean cycle time. Meanwhile, Supplier 3 has the largest.

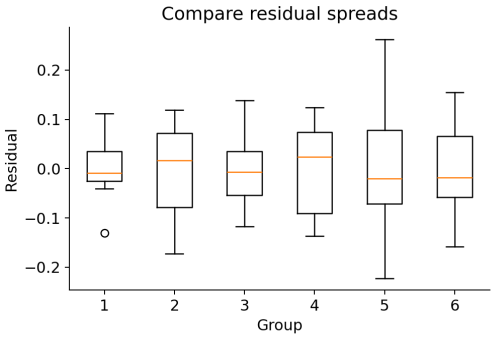
Clock ANOVA: hypotheses and assumptions

$$H_0 : \mu_1 = \cdots = \mu_6, \qquad H_a : \text{at least one mean differs.}$$

- The measured devices must be independent within and across suppliers, and this is the team's main design assumption.
- Team must check approximate normality using the residual histogram in particular.
- Team must check constant variance using residual boxplots such that it follows the Canvas MATLAB example.
- There are only 12 measurements per supplier, so the team interprets assumption problems cautiously.

Clock ANOVA: residual diagnostics





Residuals show tails and different spreads (variance ratio is 4.82). Normality and constant variance are not firmly supported in the case.

Clock ANOVA and follow up tests

$$F(5, 66) = 17.3524, \qquad p = 6.26 \times 10^{-11}.$$

- Classical ANOVA rejects equal clock cycle means at $\alpha = 0.05$.
- Equal variance and normality concerns limit the veracity and overally strength of this conclusion.
- Two sided pooled t-tests identify individual differences under the same assumptions.
- All 15 followup p-values are unadjusted and exploratory. A failure to reject does not necessarily equal equal speeds.

Clock follow up comparisons (1/2)

Suppliers	t	p (unadjusted)	Decision
1 vs. 2	-5.814	7.536e-06	Reject H_0
1 vs. 3	-11.658	6.914e-11	Reject H_0
1 vs. 4	-3.477	0.002137	Reject H_0
1 vs. 5	-2.263	0.03382	Reject H_0
1 vs. 6	-6.642	1.117e-06	Reject H_0
2 vs. 3	-4.217	0.0003554	Reject H_0
2 vs. 4	1.922	0.06772	Fail to reject H_0
2 vs. 5	1.896	0.07112	Fail to reject H_0

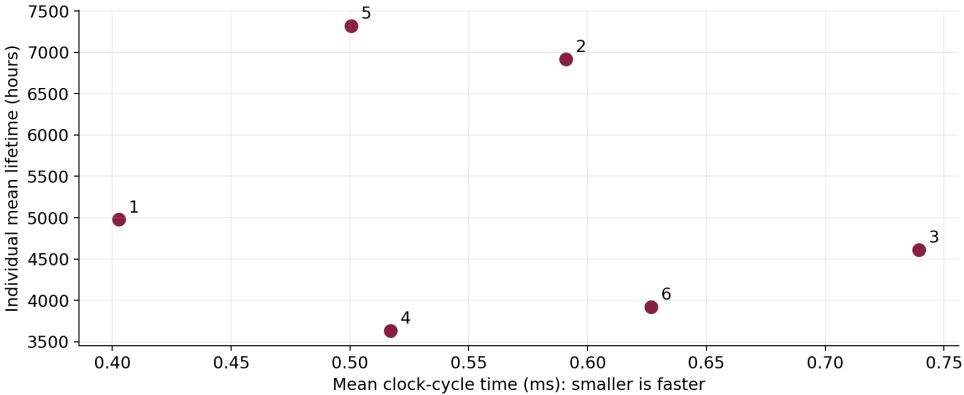
$df = 22$ for each test; assumptions remain uncertain.

Clock follow up comparisons (2/2)

Suppliers	t	p (unadjusted)	Decision
2 vs. 6	-0.912	0.3717	Fail to reject H_0
3 vs. 4	6.243	2.774e-06	Reject H_0
3 vs. 5	5.265	2.781e-05	Reject H_0
3 vs. 6	3.090	0.005354	Reject H_0
4 vs. 5	0.341	0.736	Fail to reject H_0
4 vs. 6	-2.769	0.0112	Reject H_0
5 vs. 6	-2.597	0.01645	Reject H_0

$df = 22$ for each test; assumptions remain uncertain.

Lifetime and processing speed tradeoff



Suppliers 2 and 5 lead individual mean lifetime. Supplier 1 leads observed processing speed with 5 as its closest neighbor in c-c time.

Recommendation and limits

Recommend pair 2+5 when reliability has priority

It has the largest fitted parallel MTTF (about 10,700 hours). Supplier 5's mean cycle time is 0.5006 ms; Supplier 2's is 0.5910 ms.

- If processing speed takes priority, consider 1+5: modeled MTTF about 9,350 hours, with individual mean cycle times of 0.4028 and 0.5006 ms.
- The pair's effective processing speed depends on system design, parallel clock speed gain remained unassumed.
- Another limit is that the small original samples, uncertain ANOVA assumptions, and exponential model prevent proving a uniquely best real world pair.

Methods and reproducibility

- Milestone 2: items 1, 4, and 2. This is limited to the sample summaries, exponential simulation, loops, and max lifetime for a || system.
- Program files used as resources: `large_system_reliability_ica.m`, `reliability_ica.m`, and `matlab_anova.m`.
- Specific class topics: exponential distribution, Monte Carlo, CLT, t-tests, reliability, MTTF, F-test, and ANOVA.
- MATLAB scripts are included on Canvas submission.